Fiscal Year:	FY 2021	**Task Last Updated:**	FY 10/30/2020
PI Name:	Tandon, Ritesh Ph.D.		
Project Title:	Effect of Space Radiation on Cytomegalovirus Reactivation and Lytic Replication		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Comments:			
Project Type:	Ground,New Investigation	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG. App B: Flight and Ground Space Biology Research
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	
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No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Crucian, Brian Ph.D. (NASA Johnson Space Center) Mehta, Satish Ph.D. (NASA Johnson Space Center) Simpson, Richard Ph.D. (University of Arizona)		
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Task Description:	Opportunistic pathogens such as latent viruses pose a unique hazard to crewmembers because of the ineffectiveness of preflight quarantine periods and decreased immune function. Latent herpesviruses reactivate and shed in higher quantities during and after long and short spaceflights. Human cytomegalovirus (CMV) is an important herpesvirus from public health viewpoint that causes severe systemic infections upon primary infection or on reactivation from latency. The few antiviral drugs available to treat CMV infection suffer from safety, efficacy, and resistance issues, and no vaccine against CMV is currently available. As NASA and collaborating space agencies plan for future exploration class missions (i.e., to Mars or an asteroid), eliminating the risk of an adverse health event due to alterations in virus latency, reactivation, and lytic replication is paramount. Evidence of impaired control of latent viral infections has been documented during short (i.e., 14-day Space Shuttle missions) as well as long (i.e., 6-month International Space Station missions) duration spaceflight missions, with the magnitude of this response being positively associated with time in flight. Shedding of herpesviruses including Epstein–Barr virus (EBV), varicella-zoster virus (VZV), and CMV was detected in >30% astronauts in a recent study (Mehta et al., 2017) looking at long-duration spaceflight (60–180 days). These viruses reactivated independently of each other and reactivation frequency increased in duration, and amplitude (viral copy numbers) when compared to short duration (10 to 16 days) space shuttle missions. Thus, it is easy to speculate that the substantially longer durations being proposed for exploration class missions (i.e., up to 3 years) will further increase the risk of an adverse health event due to increased virus reactivation and/or replication. Add increased radiation exposure (both duration and magnitude), limited clinical care, and the impossibility of rapidly returning a casualty back to Earth to the list of potential hazards and it is clear to see why obtaining a better understanding of virus replication under space like conditions is critical to successfully and safely send humans to, and return them from, these remote extraterrestrial destinations. In this proposal, we aim to determine the impact of space radiation on CMV latency, reactivation, and lytic replication. While most studies on this topic in the past have focused on changes in human immune system that could induce virus reactivation, we will approach this topic from both virus and host standpoints. We will study differences in CMV latency and reactivation before and after radiation exposure and determine if these changes can be correlated with specific changes in virus genome and/or transcriptome. Also, because virus control, dissemination, and resolution are ultimately governed by the competency, or lack thereof, of the host immune system, we will determine the impact of pre-exposing NK-cells and T-cells to space radiation on their ability to control active CMV infections. In this project, we will focus on a single variable: the space radiation, rather than looking at multiple variables such as microgravity, physical and psychological stresses of space travel, diet, medications, or other factors associated with spaceflight. The overarching hypothesis is that modeled space radiation will induce specific changes in CMV genome/transcriptome or immune control of infection that will induce CMV reactivation and/or increase in virus replication and pathogenicity. The specific aims to test this hypothesis are: Specific Aim 1: Determine the impact of space radiation on CMV genome and transcriptome. Specific Aim 2: Correlate radiation induced changes in CMV genome and/or transcriptome with CMV latency, reactivation, and lytic replication. Specific Aim 3: Determine the impact of space radiation on immune cell control in a mouse model of CMV infection. Reference: Mehta SK, Laudenslager ML, Stowe RP, Crucian BE, Feiveson AH, Sams CF, Pierson DL. (2017) Latent virus reactivation in astronauts on the international space station, NPJ Microgravity 3, 11.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Upon completion of this project we will gain a better understanding of how radiation impacts CMV latency, reactivation, and lytic replication. By measuring gene expression on infected cells exposed to radiation, using both traditional and cutting-edge techniques, we will be able to identify CMV genomic and transcriptomic changes that affect virus behavior. We will also determine if space radiation can impair NK-cell and T-cell function, allowing for CMV to replicate and disseminate in the host. This will allow us to determine the risk of CMV infection during spaceflight missions and gain a better understanding of the potential underpinning mechanisms. Findings from this study could lead to the development of appropriate countermeasures to ameliorate potential clinical risks associated with CMV reactivation in flight. Moreover, the methods incorporated here can be adapted to examine the effects of radiation on other relevant viral pathogens known to cause significant disease in immunocompromised people that may also pose a risk to crewmembers during spaceflight missions. Discovering the mechanisms of virus reactivation will help with the development of treatment modalities for patients on Earth.
Task Progress:	Future exploration class missions will take crews beyond the protection of the Earth's magnetosphere and the Van Allen Radiation belt. This will potentially expose the crews to greater doses of Galactic Cosmic Radiation (GCR) therefore increasing the risk of adverse health events for these crewmembers. GCR is a mixture of different radiation sources including protons, helium ions, and heavy ions coming from a myriad of galactic sources including the sun. Its particles move at nearly the speed of light and easily penetrate shuttle and space station walls, as well as human skin, cells, and DNA. It is very likely that GCR will exacerbate an already progressive CMV reactivation. CMV reactivation has both short-term and long-term consequences for the infected crewmembers as spaceflight induces immune compromise that interferes with its capacity to control viral reactivation. In the absence of immune control, a progressive CMV infection can lead to multiple end organ diseases such as chorioretinitis, encephalitis, gastroenteritis, and pneumonia (Seitz 2010). It is therefore necessary to determine the impact of space radiation on CMV latency and reactivation in order to make space travel safer. This study, therefore, sought to investigate the effect of simulated GCR on the reactivation rates of CMV in Kasumi-3 myeloid progenitor cells. Latently infected Kasumi-3 cells were treated with varying gamma radiation doses to evaluate potential differences in reactivation kinetics. Potential outcome variables were determined as changes in viral load, and changes/modifications in gene transcription. Reference: Seitz, R. (2010). Human cytomegalovirus (HCMV)-revised. Transfus. Med. Hemother. 37, 365–375.
Bibliography Type:	Description: (Last Updated: 11/21/2025)
Articles in Peer-reviewed Journals	Jin W, Zhang W, Mitra D, McCandless MG, Sharma P, Tandon R, Zhang F, Linhardt RJ. "The structure-activity relationship of the interactions of SARS-CoV-2 spike glycoproteins with glucuronomannan and sulfated galactofucan from Saccharina japonica." Int J Biol Macromol. 2020 Nov 15;163:1649-58. https://doi.org/10.1016/j.ijbiomac.2020.09.184 ; PMID: 32979436; PMCID: PMC7513770 , Nov-2020

Articles in Peer-reviewed Journals	Mishra PK, Tandon R, Byrareddy SN. "Diabetes and COVID-19 risk: A miRNA perspective." Am J Physiol Heart Circ Physiol, 2020 Sep 1;319(3):H604-H609. https://doi.org/10.1152/ajpheart.00489.2020 ; PMID: 32762561; PMCID: PMC7509271 , Sep-2020
Articles in Peer-reviewed Journals	Tandon R, Sharp JS, Zhang F, Pomin VH, Ashpole NM, Mitra D, McCandless MG, Jin W, Liu H, Sharma P, Linhardt RJ. "Effective inhibition of SARS-CoV-2 entry by heparin and enoxaparin derivatives." J Virol. 2021 Jan 13;95(3):e01987-20. https://doi.org/10.1128/JVI.01987-20 ; PubMed PMID: 33173010; PubMed Central PMCID: PMC7925120 , Jan-2021